HW5 - Econ 144

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Packages

```
library(readr)
library(lattice)
library(foreign)
library(MASS)
library(car)
```

```
## Loading required package: carData
```

```
require(stats)
require(stats4)
```

```
## Loading required package: stats4
```

```
library(KernSmooth)
```

```
## KernSmooth 2.23 loaded
## Copyright M. P. Wand 1997-2009
```

```
library(fastICA)
library(cluster)
library(leaps)
library(mgcv)
```

```
## Loading required package: nlme
```

```
## This is mgcv 1.9-1. For overview type 'help("mgcv-package")'.
```

```
library(rpart)
library(pan)
library(mgcv)
library(DAAG)
```

```
##
## Attaching package: 'DAAG'
```

```
## The following object is masked from 'package:car':
##
##     vif

## The following object is masked from 'package:MASS':
##
##     hills
```

```r
library("TTR")
library(tis)
```

```
##
## Attaching package: 'tis'

## The following object is masked from 'package:TTR':
##
##     lags

## The following object is masked from 'package:mgcv':
##
##     ti
```

```r
require("datasets")
require(graphics)
library(forecast)
```

```
## Registered S3 method overwritten by 'quantmod':
##   method            from
##   as.zoo.data.frame zoo

##
## Attaching package: 'forecast'

## The following object is masked from 'package:tis':
##
##     easter

## The following object is masked from 'package:nlme':
##
##     getResponse
```

```r
require(astsa)
```

```
## Loading required package: astsa

##
## Attaching package: 'astsa'

## The following object is masked from 'package:forecast':
##
##     gas
```

```r
library(RColorBrewer)
library(plotrix)
library(tsibble)
```

```
## Registered S3 method overwritten by 'tsibble':
##   method               from
##   as_tibble.grouped_df dplyr

##
## Attaching package: 'tsibble'

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, union
```

```r
library(ggplot2)
library(tidyr)
library(USgas)
library(fpp3)
```

```
## -- Attaching packages -------------------------------------------- fpp3 1.0.2 --

## v tibble      3.2.1     v tsibbledata 0.4.1
## v dplyr       1.1.4     v feasts      0.4.2
## v lubridate   1.9.4     v fable       0.4.1

## -- Conflicts ------------------------------------------------- fpp3_conflicts --
## x feasts::ACF()         masks nlme::ACF()
## x dplyr::between()      masks tis::between()
## x dplyr::collapse()     masks nlme::collapse()
## x lubridate::date()     masks base::date()
## x lubridate::day()      masks tis::day()
## x dplyr::filter()       masks stats::filter()
## x lubridate::hms()      masks tis::hms()
## x tsibble::intersect()  masks base::intersect()
## x lubridate::interval() masks tsibble::interval()
## x dplyr::lag()          masks stats::lag()
## x lubridate::month()    masks tis::month()
## x lubridate::period()   masks tis::period()
## x lubridate::POSIXct()  masks tis::POSIXct()
## x lubridate::quarter()  masks tis::quarter()
## x dplyr::recode()       masks car::recode()
## x dplyr::select()       masks MASS::select()
## x tsibble::setdiff()    masks base::setdiff()
## x lubridate::today()    masks tis::today()
## x tsibble::union()      masks base::union()
## x lubridate::year()     masks tis::year()
## x lubridate::ymd()      masks tis::ymd()
```

Textbook A: Forecasting for Economics and Business

Data

```r
fm <- read.csv("fmhpi_master_file.csv")
head(fm)
```

```
##   Year Month GEO_Type GEO_Name GEO_Code Index_NSA Index_SA
## 1 1975     1    State       AK        .  35.05465 35.46056
## 2 1975     2    State       AK        .  35.55635 36.03117
## 3 1975     3    State       AK        .  36.07694 36.44777
## 4 1975     4    State       AK        .  36.62437 36.72499
## 5 1975     5    State       AK        .  37.22699 36.95556
## 6 1975     6    State       AK        .  37.88329 37.24204
```

```r
table(fm$GEO_Type)
```

```
##
##   CBSA  State     US
## 235683  31059    609
```

```r
msa <- subset(fm, GEO_Type == "CBSA")
MSA1_name <- "Oxnard-Thousand Oaks-Ventura CA"
MSA2_name <- "Los Angeles-Long Beach-Anaheim CA"

msa1 <- subset(msa, GEO_Name == MSA1_name)
msa2 <- subset(msa, GEO_Name == MSA2_name)

msa1$Date <- as.Date(paste(msa1$Year, msa1$Month, "01", sep = "-"))
msa2$Date <- as.Date(paste(msa2$Year, msa2$Month, "01", sep = "-"))

msa1 <- msa1[order(msa1$Date), ]
msa2 <- msa2[order(msa2$Date), ]

head(msa1)
```

```
##        Year Month GEO_Type                        GEO_Name GEO_Code Index_NSA
## 182701 1975     1     CBSA Oxnard-Thousand Oaks-Ventura CA    37100  13.90367
## 182702 1975     2     CBSA Oxnard-Thousand Oaks-Ventura CA    37100  14.00809
## 182703 1975     3     CBSA Oxnard-Thousand Oaks-Ventura CA    37100  14.20650
## 182704 1975     4     CBSA Oxnard-Thousand Oaks-Ventura CA    37100  14.50231
## 182705 1975     5     CBSA Oxnard-Thousand Oaks-Ventura CA    37100  14.74632
## 182706 1975     6     CBSA Oxnard-Thousand Oaks-Ventura CA    37100  14.85763
##        Index_SA       Date
## 182701 13.89219 1975-01-01
## 182702 13.99511 1975-02-01
## 182703 14.14028 1975-03-01
## 182704 14.37169 1975-04-01
## 182705 14.59592 1975-05-01
## 182706 14.69720 1975-06-01
```

```r
head(msa2)
```

```
##        Year Month GEO_Type                           GEO_Name GEO_Code Index_NSA
## 140071 1975     1     CBSA Los Angeles-Long Beach-Anaheim CA    31080  15.72848
## 140072 1975     2     CBSA Los Angeles-Long Beach-Anaheim CA    31080  15.88764
## 140073 1975     3     CBSA Los Angeles-Long Beach-Anaheim CA    31080  16.15511
## 140074 1975     4     CBSA Los Angeles-Long Beach-Anaheim CA    31080  16.53437
## 140075 1975     5     CBSA Los Angeles-Long Beach-Anaheim CA    31080  16.85256
## 140076 1975     6     CBSA Los Angeles-Long Beach-Anaheim CA    31080  17.00757
##        Index_SA       Date
## 140071 15.70656 1975-01-01
## 140072 15.86993 1975-02-01
## 140073 16.08712 1975-03-01
## 140074 16.40525 1975-04-01
## 140075 16.71223 1975-05-01
## 140076 16.86140 1975-06-01
```

11.1

```r
dat <- merge(
  data.frame(
    Date = msa1$Date,
    g1   = c(NA, 100 * diff(log(msa1$Index_SA)))  # Oxnard growth
  ),
  data.frame(
    Date = msa2$Date,
    g2   = c(NA, 100 * diff(log(msa2$Index_SA)))  # LA growth
  ),
  by = "Date"
)
dat <- dat[complete.cases(dat), ]

summary(dat)
```

```
##       Date                  g1                  g2
##  Min.   :1975-02-01   Min.   :-9.64018   Min.   :-11.23512
##  1st Qu.:1987-09-23   1st Qu.: 0.01462   1st Qu.: -0.04954
##  Median :2000-05-16   Median : 0.55311   Median :  0.59246
##  Mean   :2000-05-16   Mean   : 0.51802   Mean   :  0.53293
##  3rd Qu.:2013-01-08   3rd Qu.: 1.10792   3rd Qu.:  1.15150
##  Max.   :2025-09-01   Max.   : 9.26768   Max.   : 10.79841
```

```r
cor(dat$g1, dat$g2)
```

```
## [1] 0.9617833
```

```r
dat$Month <- yearmonth(dat$Date)
dat_ts <- as_tsibble(dat, index = Month)
```

```
fit <- model(
  dat_ts,
  var = VAR(vars(g1, g2) ~ AR(p = 1:6), ic = "bic")
)

report(fit)
```

```
## Series: g1, g2
## Model: VAR(6)
##
## Coefficients for g1:
##       lag(g1,1)  lag(g2,1)  lag(g1,2)  lag(g2,2)  lag(g1,3)  lag(g2,3)
##         -0.3555     1.7797     3.9526    -5.4028    -7.9026     9.3101
## s.e.     0.6750     0.6627     1.8120     1.7802     2.4177     2.3830
##       lag(g1,4)  lag(g2,4)  lag(g1,5)  lag(g2,5)  lag(g1,6)  lag(g2,6)
##         10.8388   -11.8567    -9.8039    10.5326     4.2191    -4.3851
## s.e.     2.4097     2.3788     1.7788     1.7575     0.6521     0.6447
##
## Coefficients for g2:
##       lag(g1,1)  lag(g2,1)  lag(g1,2)  lag(g2,2)  lag(g1,3)  lag(g2,3)
##         -2.9141     4.3111     6.5283    -7.9594    -9.3905    10.7901
## s.e.     0.6876     0.6750     1.8457     1.8134     2.4628     2.4274
##       lag(g1,4)  lag(g2,4)  lag(g1,5)  lag(g2,5)  lag(g1,6)  lag(g2,6)
##         11.6765   -12.6828   -10.3300    11.0552     4.4139    -4.5736
## s.e.     2.4546     2.4232     1.8119     1.7903     0.6643     0.6567
##
## Residual covariance matrix:
##        g1     g2
## g1 0.2295 0.2333
## g2 0.2333 0.2381
##
## log likelihood = 872.97
## AIC = -1689.95   AICc = -1687.11 BIC = -1566.74
```

The two samples used are the MSA rates in Oxnard and LA. They both have 609 observations, which were taken monthly since 1975.

Looking at the above data (descriptive and forecasting), we can see that the growth rate in MSA1 (Oxnard) is 0.52%, very similar to that of MSA2 (LA), which is 0.53%. They two sets are very correlated (0.96).

Using the fpp3 package's var() function, chosen using BIC to keep models simple, we find that a var(6) model leads to the most accurate results. BIC is minimized at p=6 which is why var(6) was chosen. In this model, almost all of the lag coefficients are significant, implying a strong correlation between prior lags and current lags. There are also clear ties between Oxnard rates and LA rates.

11.2

```
p <- 6
n <- nrow(dat)
idx <- (p + 1):n

Y1 <- dat$g1[idx]  # Oxnard_t
```

```r
Y2 <- dat$g2[idx]  # LA_t

X <- NULL
for (i in 1:p) {
  X <- cbind(X, dat$g1[idx - i], dat$g2[idx - i])
}
X <- as.data.frame(X)
names(X) <- unlist(lapply(1:p, function(i) c(paste0("g1_L", i), paste0("g2_L", i))))

df <- data.frame(Y1 = Y1, Y2 = Y2, X)

fit1_u <- lm(Y1 ~ ., data = df)
fit2_u <- lm(Y2 ~ ., data = df)

cols_g1 <- grep("^g1_L", names(df), value = TRUE)
cols_g2 <- grep("^g2_L", names(df), value = TRUE)

fit1_r <- lm(Y1 ~ ., data = df[, c("Y1", cols_g1), drop = FALSE])
fit2_r <- lm(Y2 ~ ., data = df[, c("Y2", cols_g2), drop = FALSE])

gc_g2_to_g1 <- anova(fit1_r, fit1_u)
gc_g1_to_g2 <- anova(fit2_r, fit2_u)

gc_g2_to_g1
```

```
## Analysis of Variance Table
##
## Model 1: Y1 ~ g1_L1 + g1_L2 + g1_L3 + g1_L4 + g1_L5 + g1_L6
## Model 2: Y1 ~ Y2 + g1_L1 + g2_L1 + g1_L2 + g2_L2 + g1_L3 + g2_L3 + g1_L4 +
##     g2_L4 + g1_L5 + g2_L5 + g1_L6 + g2_L6
##   Res.Df     RSS Df Sum of Sq     F    Pr(>F)
## 1    595 152.162
## 2    588   0.486  7    151.68 26203 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
gc_g1_to_g2
```

```
## Analysis of Variance Table
##
## Model 1: Y2 ~ g2_L1 + g2_L2 + g2_L3 + g2_L4 + g2_L5 + g2_L6
## Model 2: Y2 ~ Y1 + g1_L1 + g2_L1 + g1_L2 + g2_L2 + g1_L3 + g2_L3 + g1_L4 +
##     g2_L4 + g1_L5 + g2_L5 + g1_L6 + g2_L6
##   Res.Df     RSS Df Sum of Sq     F    Pr(>F)
## 1    595 168.319
## 2    588   0.504  7    167.81 27947 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

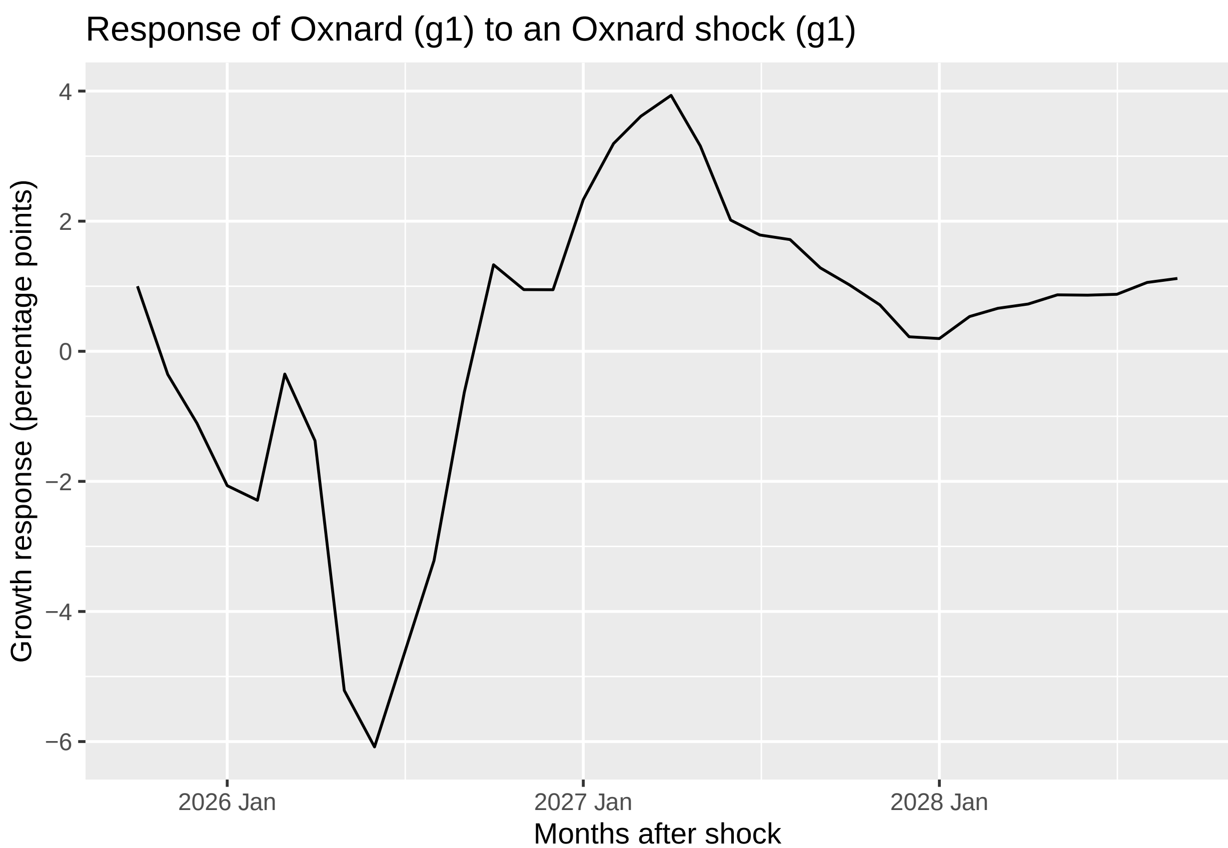
Looking at the Granger Causality results, we see that regressing y2 on y1 and y1 on y2 both yeild significant p values (sub 5%). This means that LA Granger-causes Oxnard MSA and Oxnard Granger-causes LA MSA. In both cases, we can reject the "no cause" null hypothesis.

11.3

```r
h <- 36
future <- new_data(dat_ts, n = h)

irf_g1 <- IRF(fit, new_data = future, impulse = "g1")

irf_g2 <- IRF(fit, new_data = future, impulse = "g2")



autoplot(irf_g1, vars = vars(g1)) +
  labs(
    title = "Response of Oxnard (g1) to an Oxnard shock (g1)",
    x = "Months after shock",
    y = "Growth response (percentage points)"
  )
```

```
## Plot variable not specified, automatically selected `.vars = g1`
```

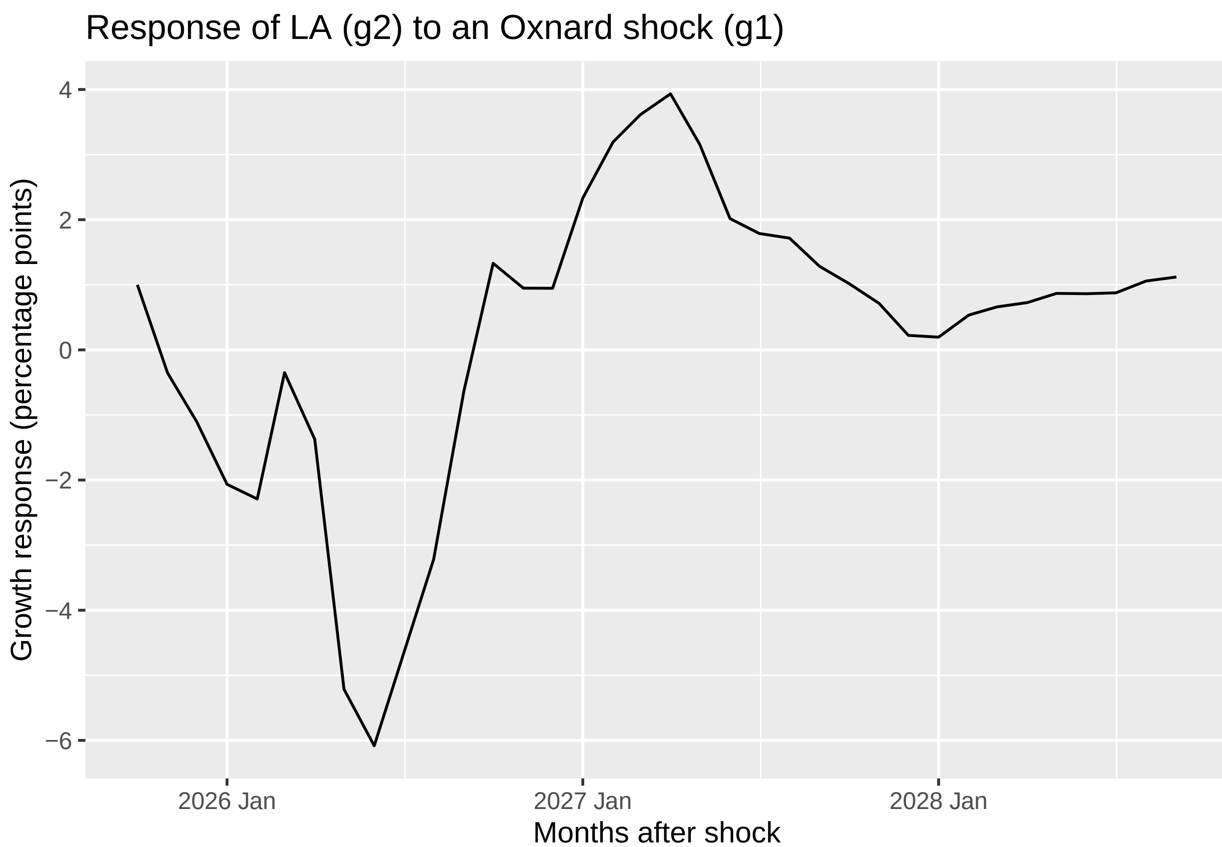
```
## Warning in geom_line(...): Ignoring unknown parameters: `vars`
```



```r
autoplot(irf_g1, vars = vars(g2)) +
  labs(
    title = "Response of LA (g2) to an Oxnard shock (g1)",
    x = "Months after shock",
    y = "Growth response (percentage points)"
  )
```

```
## Plot variable not specified, automatically selected ‘.vars = g1‘
```

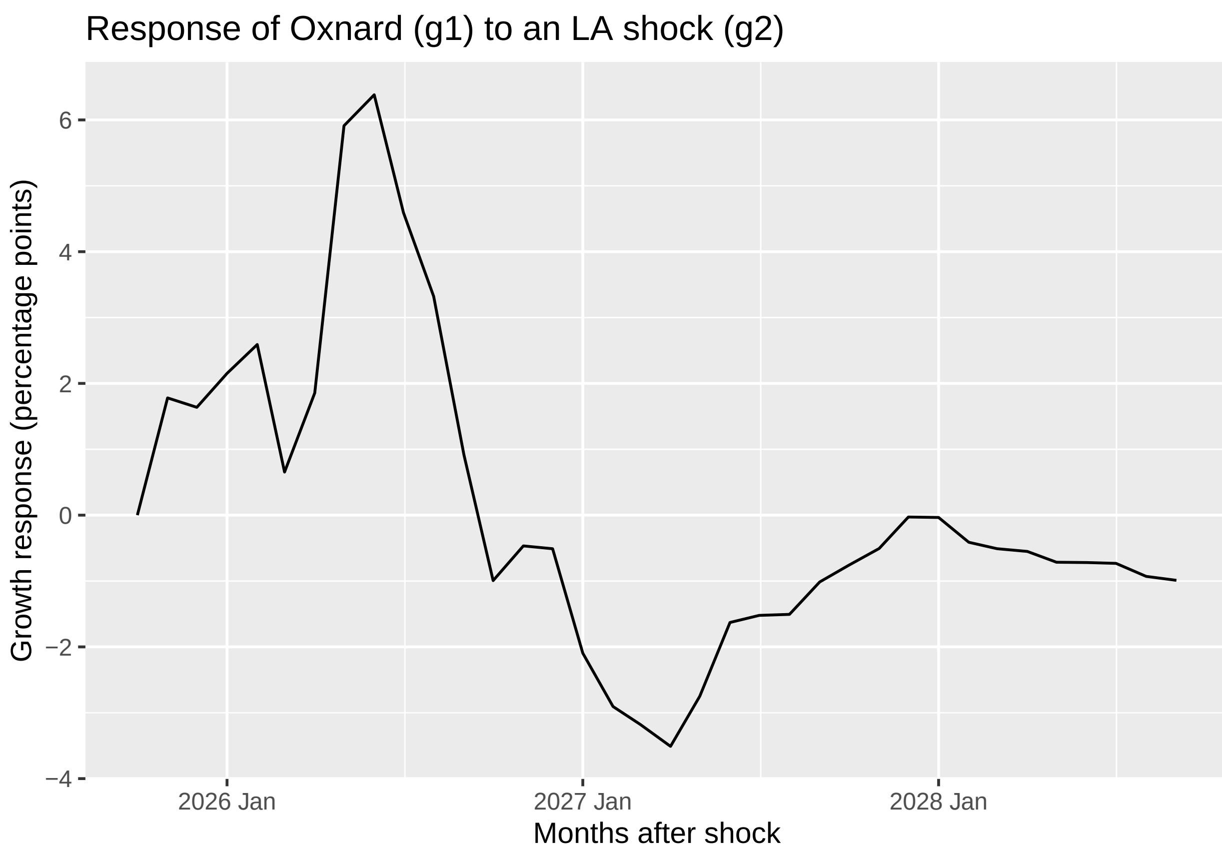
```
## Warning in geom_line(...): Ignoring unknown parameters: ‘vars‘
```




```r
autoplot(irf_g2, vars = vars(g1)) +
  labs(
    title = "Response of Oxnard (g1) to an LA shock (g2)",
    x = "Months after shock",
    y = "Growth response (percentage points)"
  )
```

```
## Plot variable not specified, automatically selected ‘.vars = g1‘
```

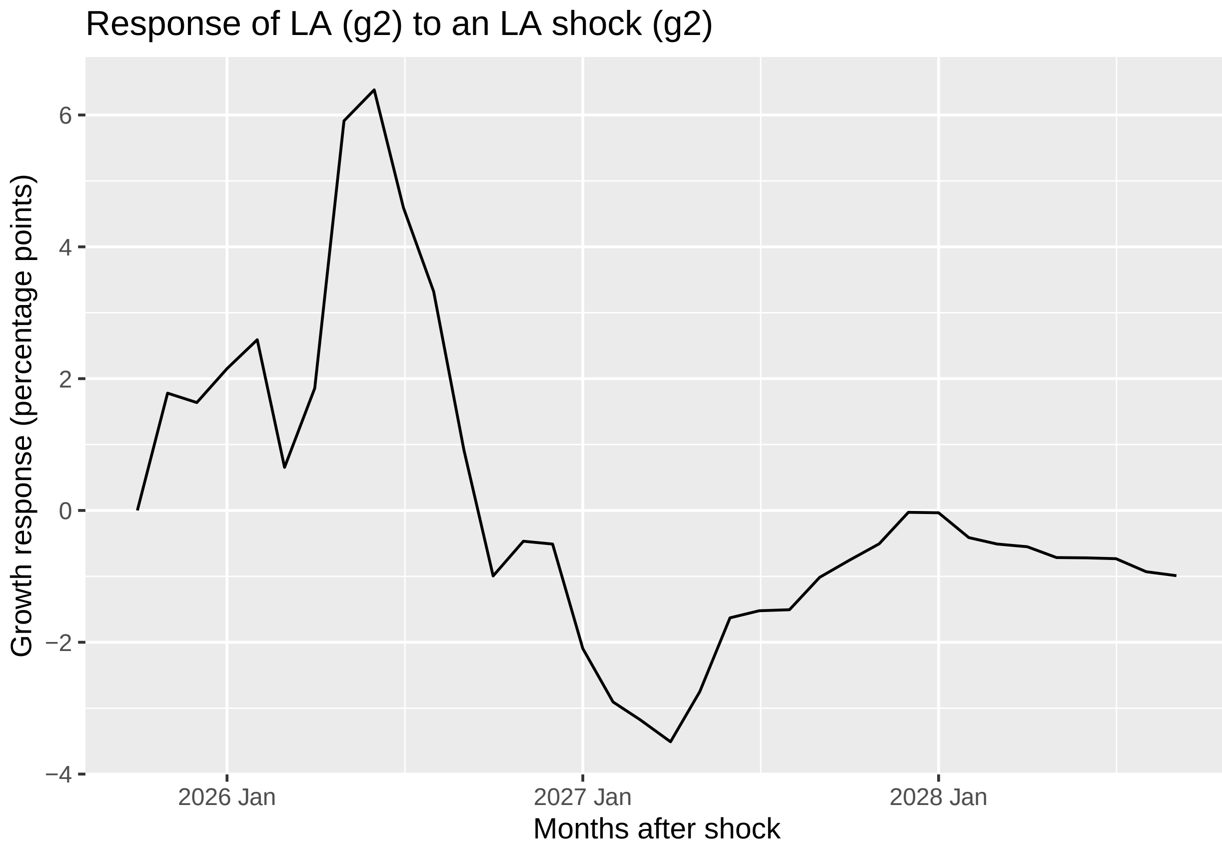
```
## Warning in geom_line(...): Ignoring unknown parameters: ‘vars‘
```




```r
autoplot(irf_g2, vars = vars(g2)) +
  labs(
    title = "Response of LA (g2) to an LA shock (g2)",
    x = "Months after shock",
    y = "Growth response (percentage points)"
  )
```

```
## Plot variable not specified, automatically selected ‘.vars = g1‘
```

```
## Warning in geom_line(...): Ignoring unknown parameters: ‘vars‘
```



The above graphs show that in forecasted data, self shocks (LA responding to LA shock, Oxnard responding to Oxnard shock) tend to be positive roughly 6 months later before a slight dip after a year. The counter shocks have an opposite effect, leading to a negative decline followed by an upturn once the self shocks begin to self correct.

Given my prior knowledge of the regions (growing up in LA and then moving to over the summer Oxnard), these results seem in line. I was surprised to see such similar growth rates since 1975, but the effects of shocks make sense to me. When LA grows, competing markets shrink, and vice versa.